

EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

PHASE 6: Project Testing Phase

1. Introduction

Testing verifies whether EduGenie performs the functions defined during requirement analysis and design. It also helps identify errors, usability problems, and unexpected behaviour before final demonstration.

2. Testing Objectives

The main objectives are to verify input handling, AI response generation, response display, interface usability, navigation, and basic error handling.

3. Functional Testing

Functional tests check whether each major feature works as intended. The team enters questions, submits them, verifies that processing occurs, and checks whether a response is displayed.

4. Question-Based Testing

Different categories of questions can be tested, including simple factual questions, concept explanations, short questions, and subject-related questions. This helps observe how the system responds to different inputs.

5. Input Validation Testing

The team checks what happens when the input is empty, too short, unclear, or unusual. The system should handle these cases without crashing or producing confusing interface behaviour.

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PHASE 6: Project Testing Phase – CONTINUED

6. AI Response Testing

Generated responses are checked for relevance, clarity, and usefulness. Since generative AI can sometimes produce incorrect or incomplete information, important answers should be verified against reliable sources.

7. Usability Testing

Students or team members can use the interface and provide feedback about navigation, readability, ease of entering questions, and understanding the responses.

8. Error Testing

Possible network failures, unavailable services, integration errors, and unexpected responses are considered. The application should provide understandable feedback where possible.

9. Corrections and Retesting

Issues found during testing are corrected and the affected functions are tested again. Retesting confirms whether the correction solved the problem without creating another issue.

10. Conclusion

Testing improves the quality and reliability of EduGenie. It confirms the main workflow and identifies areas that require correction before the final project demonstration.